



**BITS Pilani**  
Pilani | Dubai | Goa | Hyderabad

Computer Vision AIML\* ZG525  
2025-26 Second Semester, M.Tech (AIML)

## Session 1: Introduction to Computer Vision

Raghavendra Singh



## **Course Objectives**

CO1: Students should understand the fundamentals of a camera producing an image, including camera calibration, optical distortions, perspective corrections etc.

CO2: Students should be familiar with various building block algorithms in Computer Vision, including Image processing and Deep Learning with emphasis on the algorithm building blocks.

CO3: Students should create at least one end-user application.



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CENGAGE

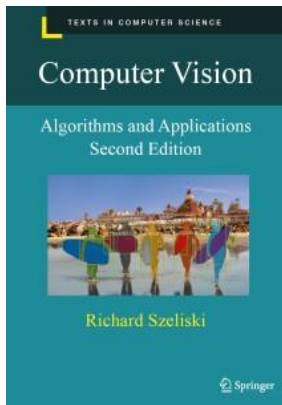
HINDUSTAN

IMAGE PROCESSING,  
ANALYSIS, & MACHINE VISION  
Milan Sonka, Vaclav Hlavac, Roger Boyle



## Textbook-1:

Image Processing, Analysis, and Machine Vision: Milan Sonka, Vaclav Hlavac, Roger Boyle, Fourth edition, Cengage Learning



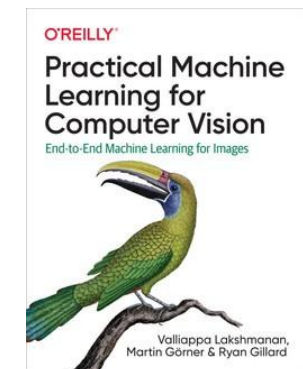
## Reference-1:

Rick Szeliski, *Computer Vision: Algorithms and Applications*

Online at: <http://szeliski.org/Book/>

## Reference-2:

*Practical Machine Learning for Computer Vision:  
End-to-End Machine Learning for Images*, O'Reilly, 2021





# Evaluation Plan

- Evaluation

Two Quizzes for 5% each; Best one towards final grading; No Makeup; → 5%

Two Assignments –  $x\% + y\% = 25\%$

Mid Term Exam - 30%

Comprehensive Exam - 40%

- Webinars/Tutorials

4 tutorials : 2 before mid-sem & 2 after mid-sem

- Teaching Assistant: TBD



# What will you learn in this course?

1. Computer Vision Fundamentals; (~2 Sessions )
2. Selected Topics in Low Level Vision & Mid Level Vision; (~3 Sessions)
3. Review of Deep Learning Approaches for Computer Vision ( ~ 1 Session) [ As Webinar ]
4. Image Classification – Problem, Deep Learning Architectures, Metrics, Use cases. ( ~2 Sessions)
5. Classic & Modern Approaches, Applications for
  - (a) Object Detection, Recognition (~2 Sessions )
  - (b) Face Detection, Recognition ( ~1 Session ) [ As Webinar ]
  - (c) Object Tracking (~2 Sessions)
  - (d) Object Segmentation ( ~2 Sessions)
  - (e) OCR (~ 1 Session) [ As Webinar ]
6. Visual Bag of Words & Semantic Hierarchy (~1 Session)
7. Edge Devices for Computer Vision (~1 Session ) [ As Webinar ]



# What will you learn in this course?

- ☐ On your learning
  - ☐ Reading materials given at the end of each class
  - ☐ Most of the topics needs to balance depth and breadth. We will use research articles and other online materials quite frequently.
  - ☐ Python demonstrations will be part of regular classes as required  $\frac{3}{4}$  TA's will join the lectures sessions to support with demonstration.
  - ☐ Wherever necessary, stress will be given to related classic topics in Computer Vision.
  - ☐ Assignment problems [to be solved in a group of maximum of 4 members ] will be given on natural images and remote sensing images & you can make a choice.



- Plagiarism Policy

All submissions for graded components must be the result of your original effort. It is strictly prohibited to copy and paste verbatim from any sources, whether online or from your peers. The use of unauthorized sources or materials, as well as collusion or unauthorized collaboration to gain an unfair advantage, is also strictly prohibited.

Please note that we will not distinguish between the person sharing their resources and the one receiving them for plagiarism, and the consequences will apply to both parties equally.

In cases where suspicious circumstances arise, such as identical verbatim answers or a significant overlap of unreasonable similarities in a set of submissions, will be investigated, and severe punishments will be imposed on all those found guilty of plagiarism.



# Agenda

Introduction to the course:

What is Computer Vision?

Why Computer Vision is hard?

Applications of Computer Vision

Image representation and image analysis tasks



# What is Computer Vision?

- **Computer vision** is the science and technology of machines that see.
- Concerned with the theory for building artificial systems that obtain information from images.
- The image data can take many forms, such as a video sequence, depth images, views from multiple cameras, or multi-dimensional data from a medical scanner



# Computer Vision

Make computers understand images and videos.



What kind of scene?

Where are the cars?

How far is the building?

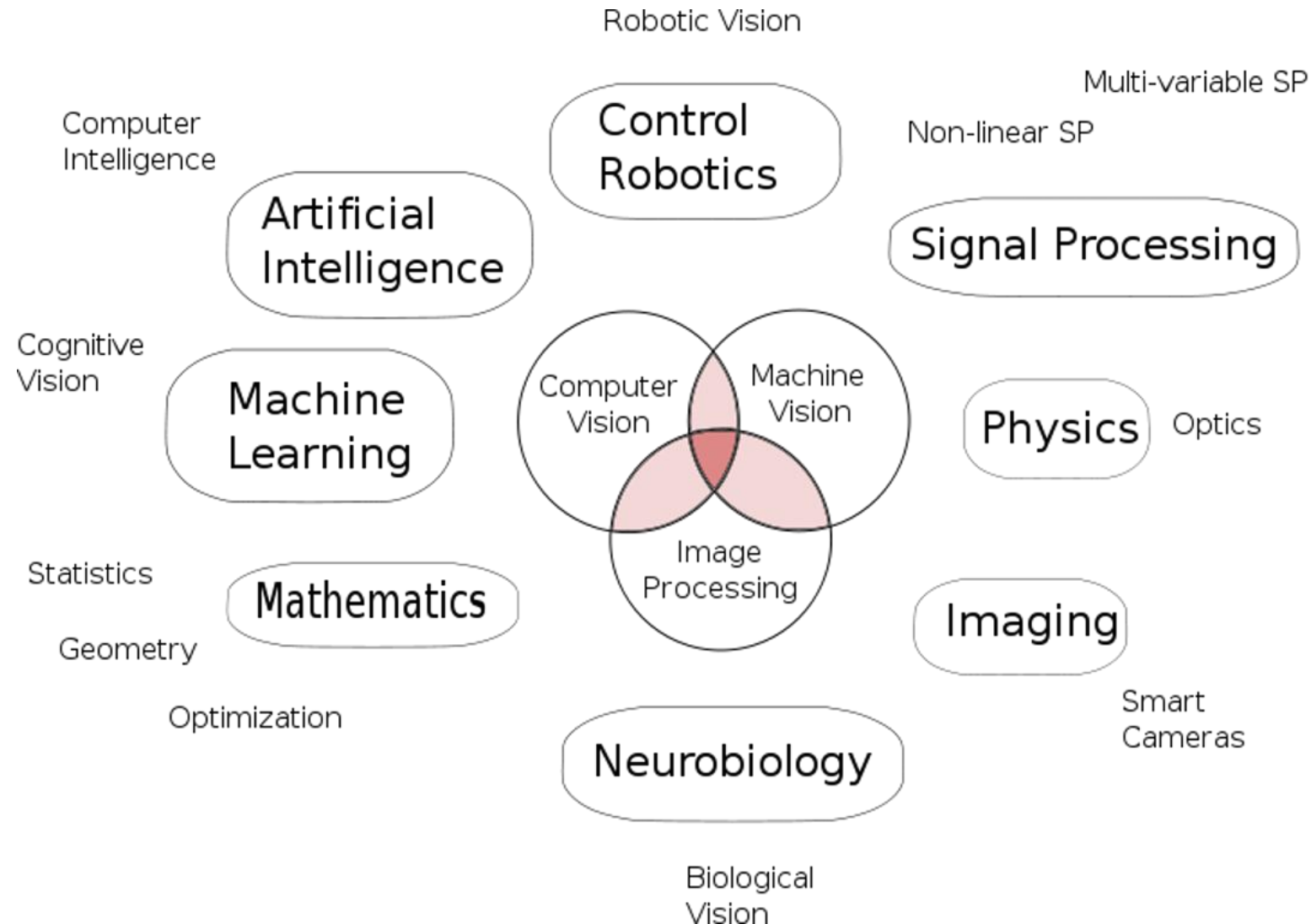
...

# Vision is really hard

- Vision is an amazing feat of natural intelligence
  - Visual cortex occupies about 50% of Macaque brain
  - More human brain devoted to vision than anything else



# Vision is Multi-disciplinary





# Every image tells a story



- Goal of computer vision: perceive the “story” behind the picture
- Compute properties of the world
  - 3D shape
  - Names of people or objects
  - What happened?

# The goal of computer vision



|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 0 | 3 | 2 | 5 | 4 | 7 | 6 | 9 | 8 |
| 3 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 2 | 1 | 0 | 3 | 2 | 5 | 4 | 7 | 6 |
| 5 | 2 | 3 | 0 | 1 | 2 | 3 | 4 | 5 |
| 4 | 3 | 2 | 1 | 0 | 3 | 2 | 5 | 4 |
| 7 | 4 | 5 | 2 | 3 | 0 | 1 | 2 | 3 |
| 6 | 5 | 4 | 3 | 2 | 1 | 0 | 3 | 2 |
| 9 | 6 | 7 | 4 | 5 | 2 | 3 | 0 | 1 |
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |

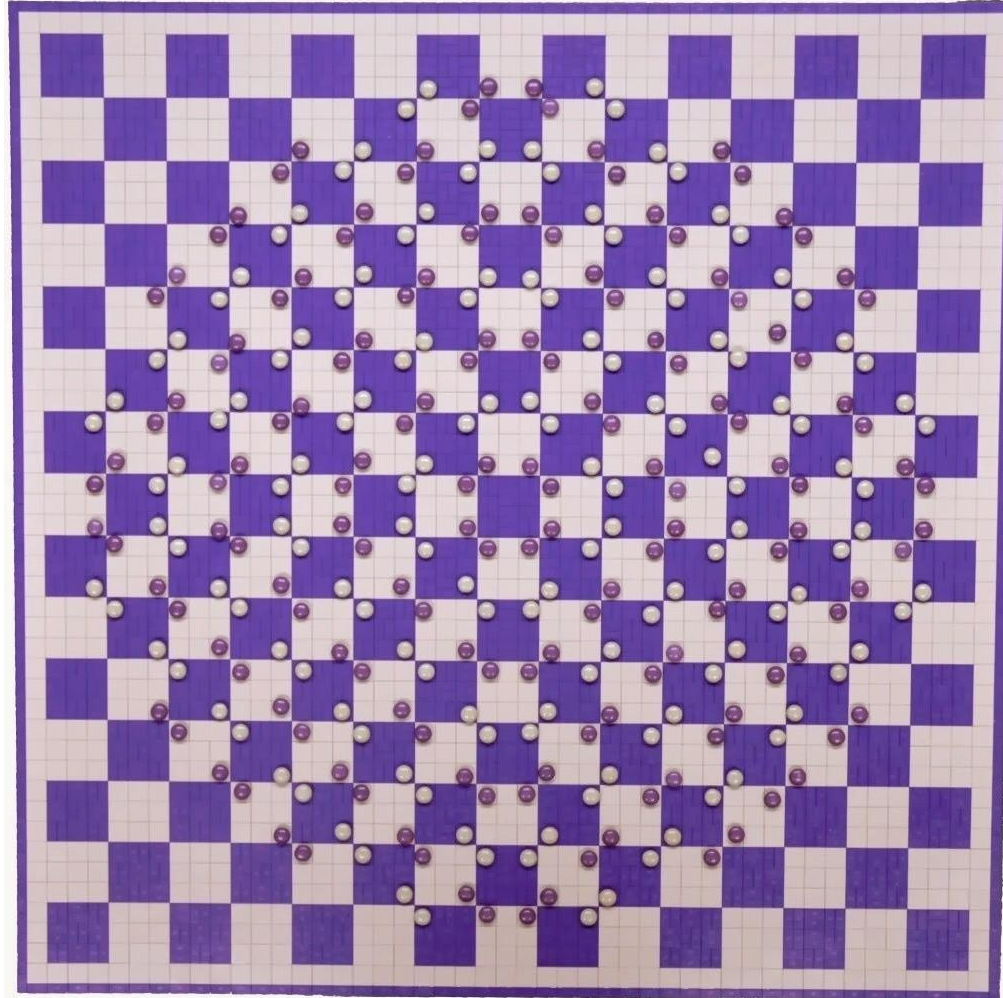
# Can computers match human perception?



- Yes and no (mainly no)
  - computers can be better at “easy” things
  - humans are better at “hard” things
- But huge progress
  - Accelerating in the last five years due to deep learning
  - What is considered “hard” keeps changing



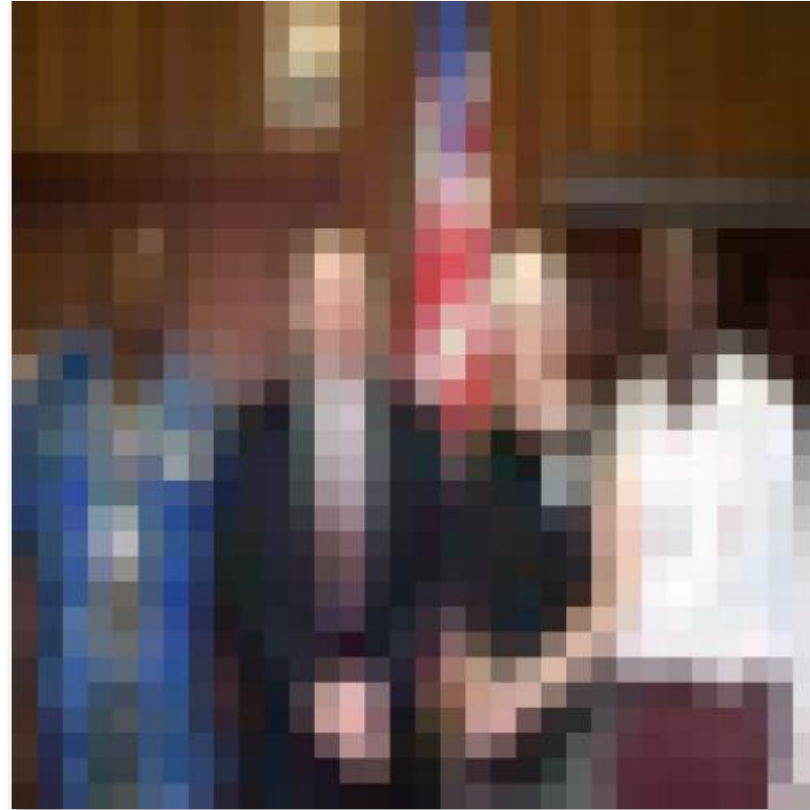
# Human perception has its shortcomings



<https://twitter.com/pickover/status/1460275132958662657/>



# But humans can tell a lot about a scene from a little information...



Source: "80 million tiny images" by Torralba, et al.



# What is in here?



# The goal of computer vision

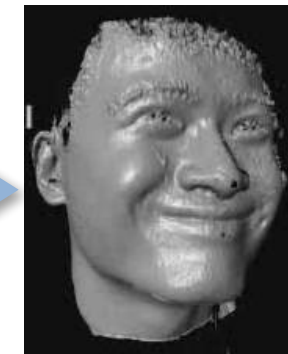
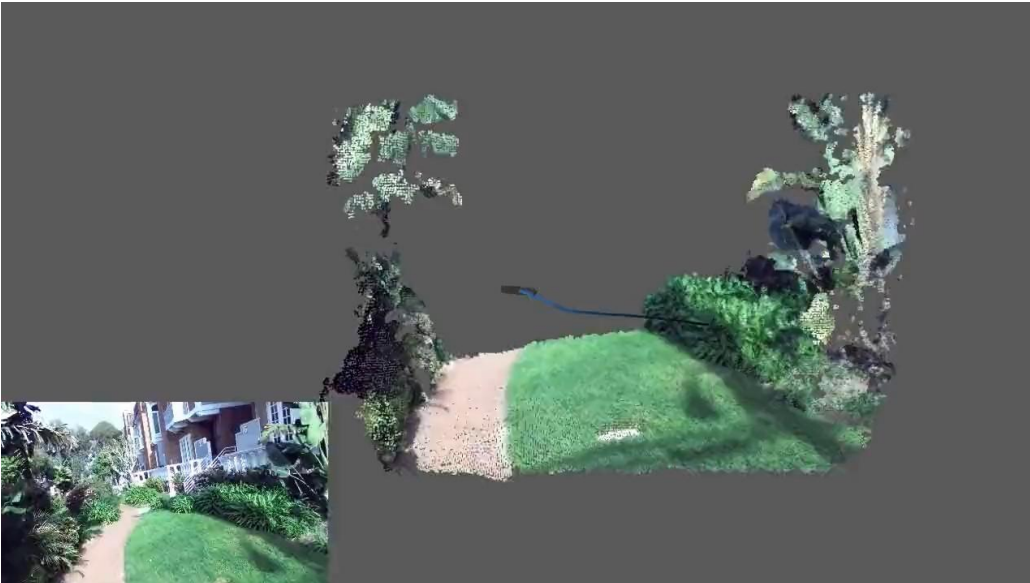
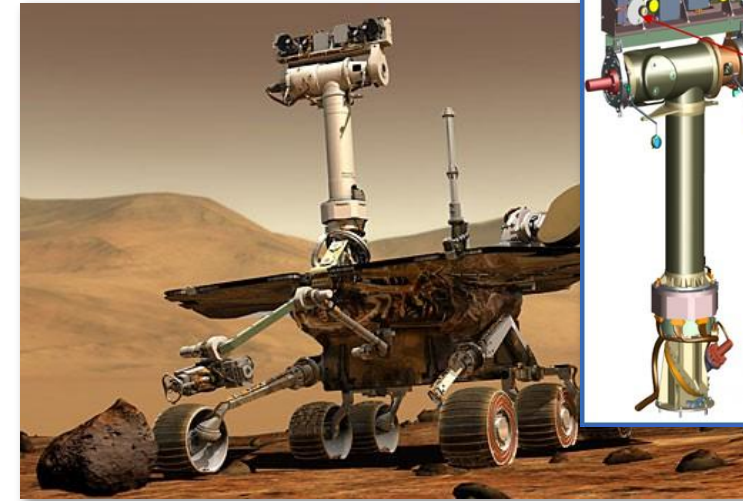


# The goal of computer vision

- Compute the 3D shape of the world



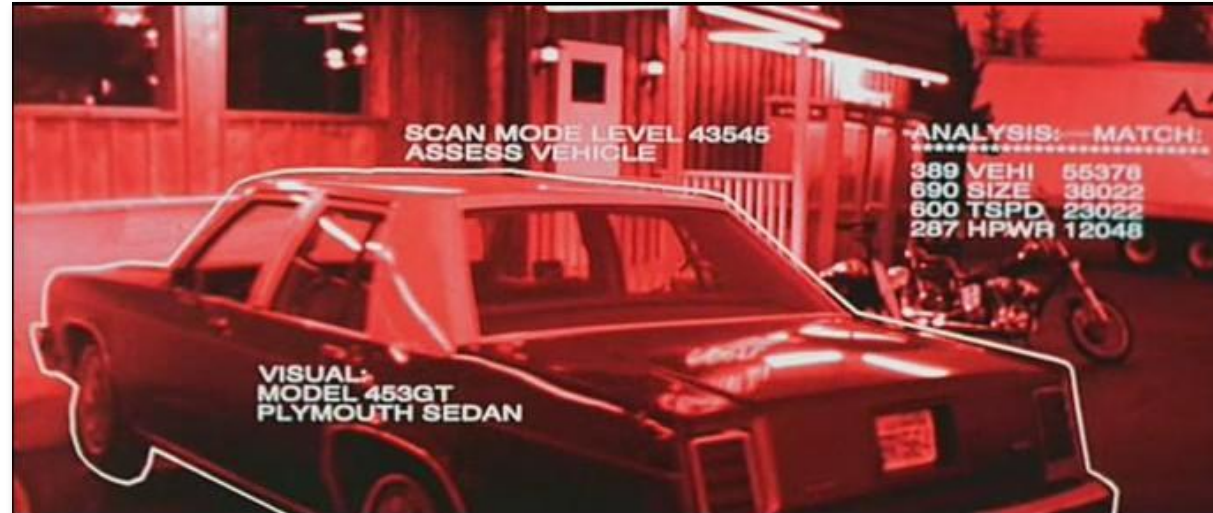
ZED 2i Camera





# The goal of computer vision

- Recognize objects and people



*Terminator 2,*  
1991







slide credit: Fei-Fei, Fergus & Torralba

# The goal of computer vision

- “Enhance” images





# The goal of computer vision

- Improve photos (“Computational Photography”)



Super-resolution (source: 2d3)



Low-light photography  
(credit: [Hasinoff et al., SIGGRAPH ASIA 2016](#))



Depth of field on cell phone camera  
(source: [Google Research Blog](#))



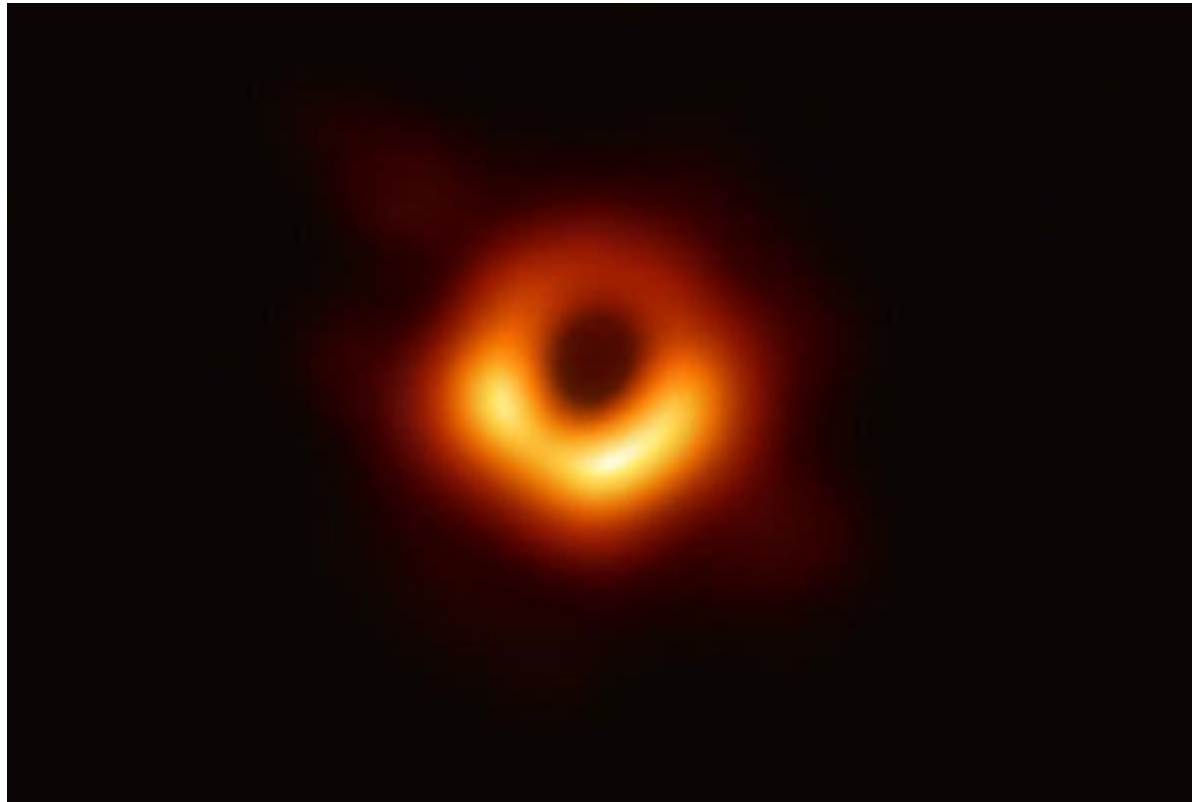
Removing objects  
([Google Magic Eraser](#))



## ***Darkness Visible, Finally: Astronomers Capture First Ever Image of a Black Hole***

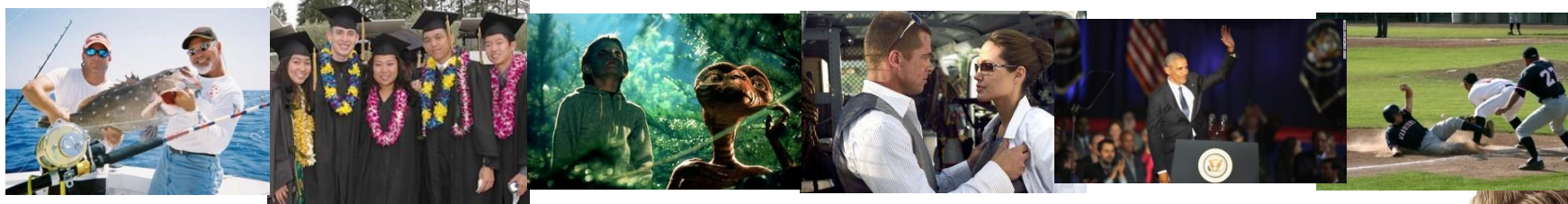
Astronomers at last have captured a picture of one of the most secretive entities in the cosmos.

April 10, 2019



# Why study computer vision?

- Billions of images/videos captured per day

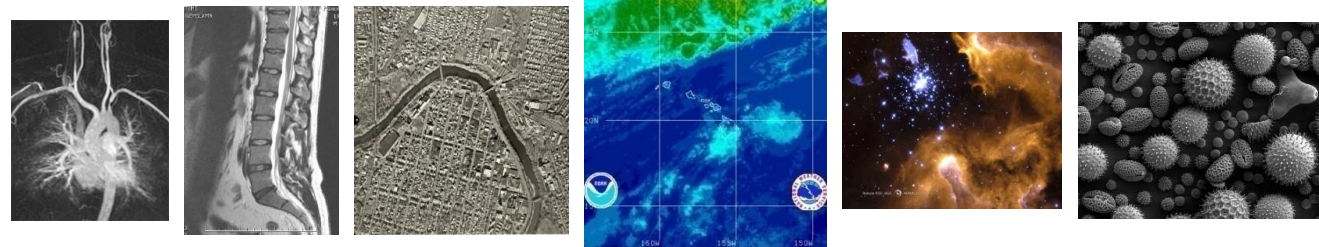


flickr



YouTube

TikTok

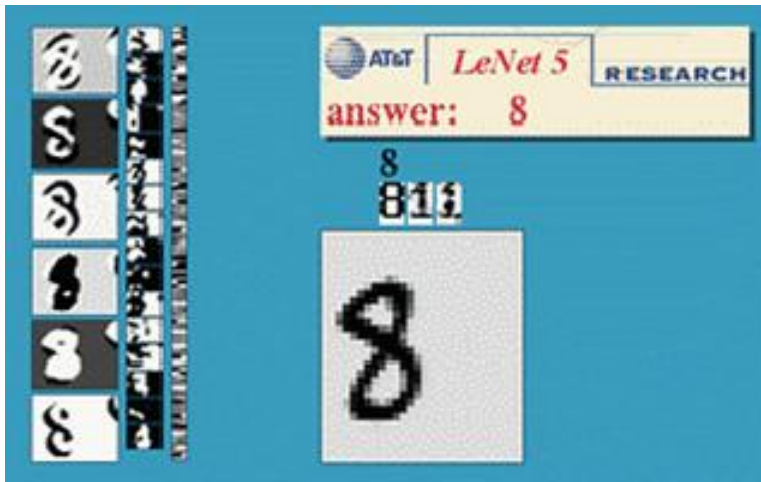


- Huge number of potential applications



# Optical character recognition (OCR)

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs  
(1990's)

<http://yann.lecun.com/exdb/lenet/>

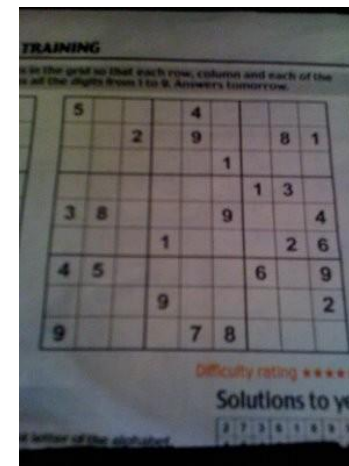


Automatic check



License plate readers

[http://en.wikipedia.org/wiki/Automatic\\_number\\_plate\\_recognition](http://en.wikipedia.org/wiki/Automatic_number_plate_recognition)



Sudoku grabber

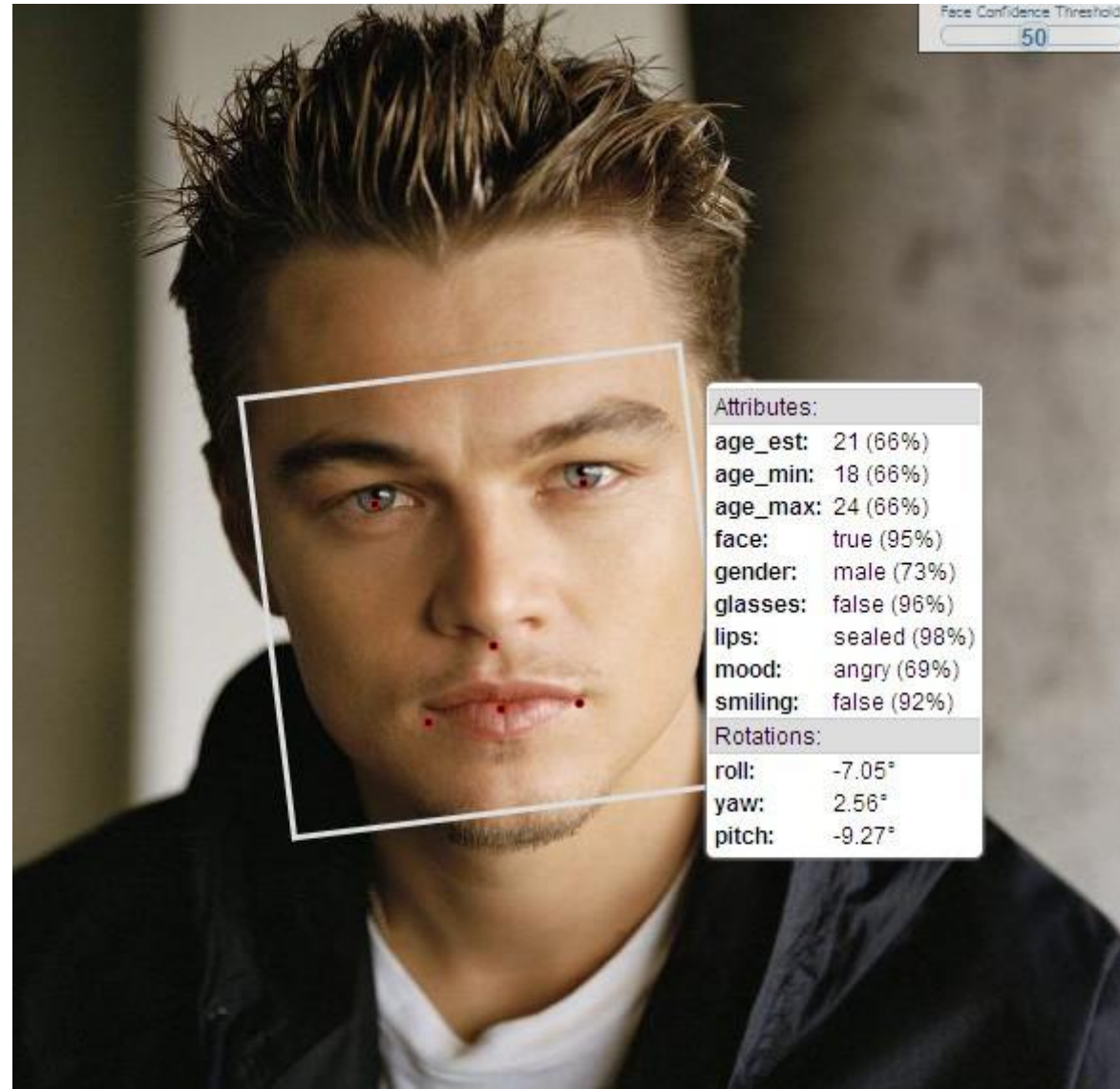
<http://sudokugrab.blogspot.com/>

# Face detection



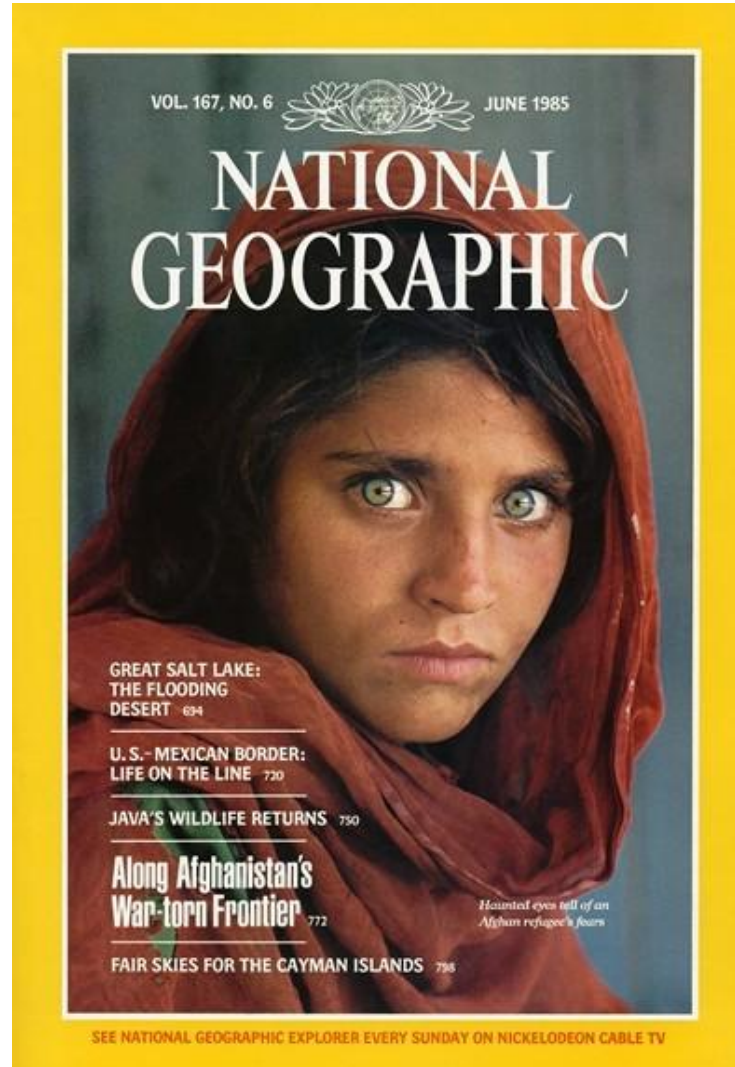
- Nearly all cameras detect faces in real time
  - (Why?)

# Face analysis and recognition





# Vision-based biometrics



Who is she?

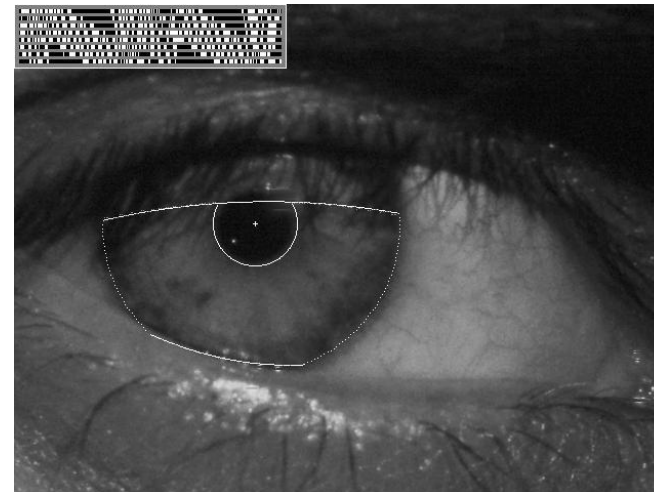
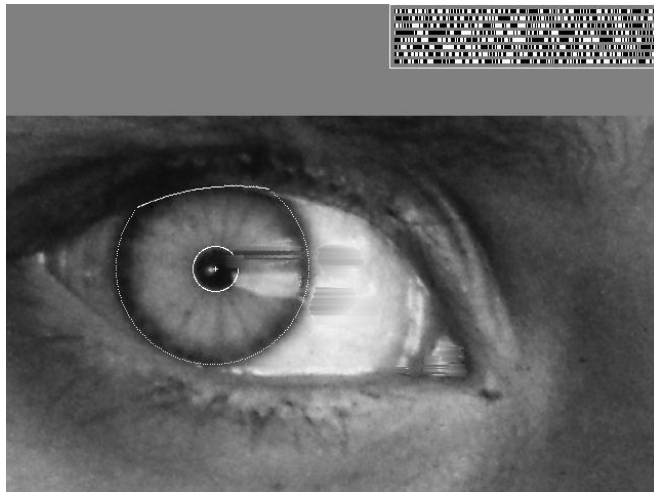
Source: S.  
Seitz



# Vision-based biometrics

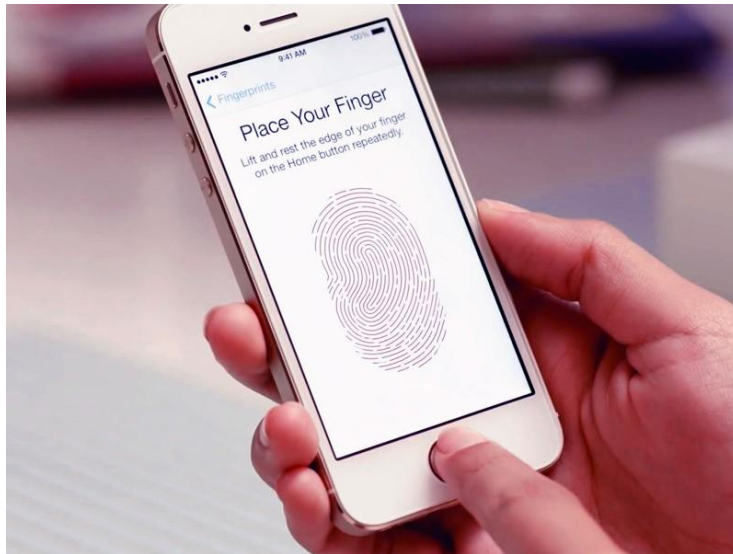


*“How the Afghan Girl was Identified by Her Iris Patterns”* [ [Story](#) , [youtube link](#) ]



Source: S.  
Seitz

# Login without a password



Fingerprint scanners on many new smartphones and other devices



Face unlock on Apple iPhone X  
See also <http://www.sensiblevision.com/>



# Bird identification



Merlin Bird ID (based on Cornell Tech technology!)

# Special effects: shape capture



*The Matrix* movies, ESC Entertainment, XYZRGB, NRC

Source: S. Seitz



# Special effects: motion capture



*Pirates of the Caribbean*, Industrial Light and Magic

Source: S. Seitz

# 3D face tracking w/ consumer cameras



Snapchat Lenses



[Face2Face system](#) (Thies et al.)

# Image synthesis



Karras, et al., *Progressive Growing of GANs for Improved Quality, Stability, and Variation*, ICLR 2018



# Image synthesis



“An astronaut riding a horse in a photorealistic style” – DALL-E 2

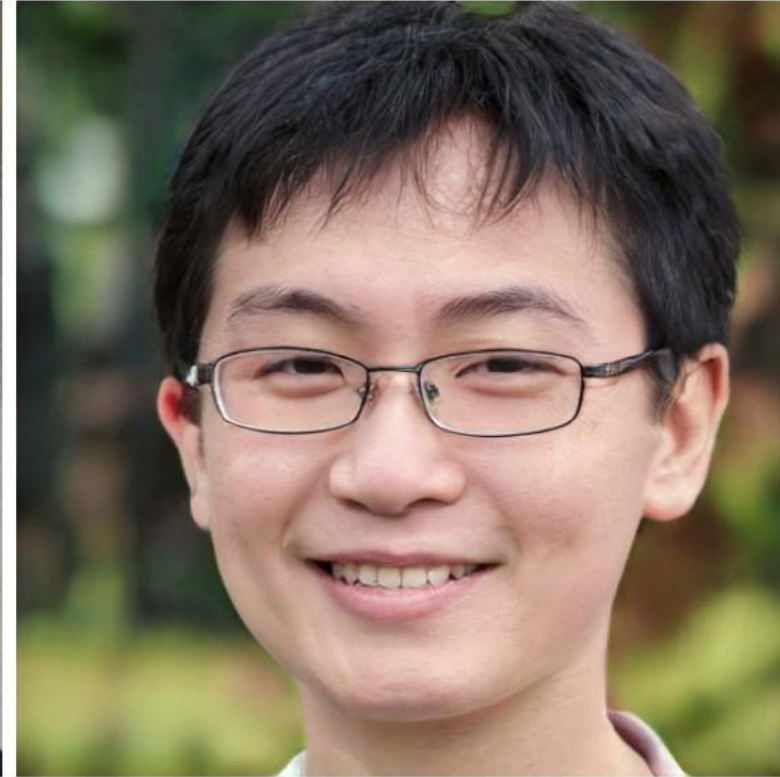


“A photo of a Corgi dog riding a bike in Times Square. It is wearing sunglasses and a beach hat”  
– Imagen



# Which face is real?

Click on the person who is real.



<https://www.whichfaceisreal.com/>

# Smart cars



The screenshot displays the Mobileye website with a top navigation bar containing "manufacturer products" and "consumer products". The main banner features a car with four camera fields of view: "rear looking camera", "forward looking camera", and "side looking camera", under the heading "Our Vision. Your Safety.".

Below the banner are three product sections:

- EyeQ Vision on a Chip**: Shows an EyeQ chip with a "read more" link.
- Vision Applications**: Describes "Road, Vehicle, Pedestrian Protection and more" with an image of a pedestrian and a "read more" link.
- AWS Advance Warning System**: Shows a dashboard display with a car icon and "0.8" and a "read more" link.

On the right sidebar, the "News" section lists:

- Mobileye Advanced Technologies Power Volvo Cars World First Collision Warning With Auto Brake System
- Volvo: New Collision Warning with Auto Brake Helps Prevent Rear-end

The "Events" section lists:

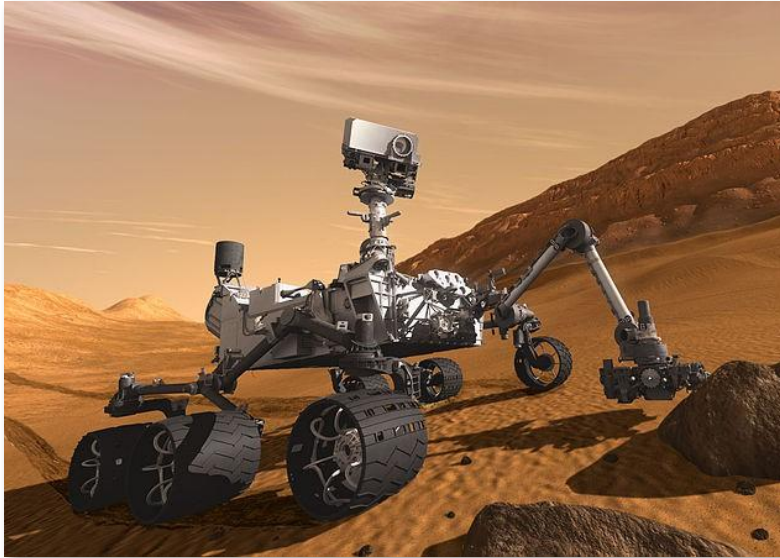
- Mobileye at Equip Auto, Paris, France
- Mobileye at SEMA, Las Vegas, NV

Both sidebar sections include a "read more" link.

- [Mobileye](#)
- Tesla Autopilot
- Safety features in many cars



# Robotics



NASA's Mars Curiosity Rover

[https://en.wikipedia.org/wiki/Curiosity\\_\(rover\)](https://en.wikipedia.org/wiki/Curiosity_(rover))



Amazon Picking Challenge

<http://www.robocup2016.org/en/events/amazon-picking-challenge/>



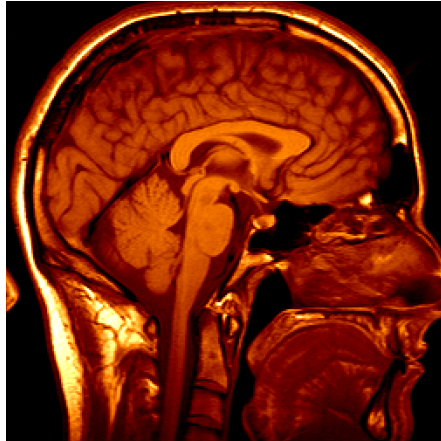
Amazon Prime Air



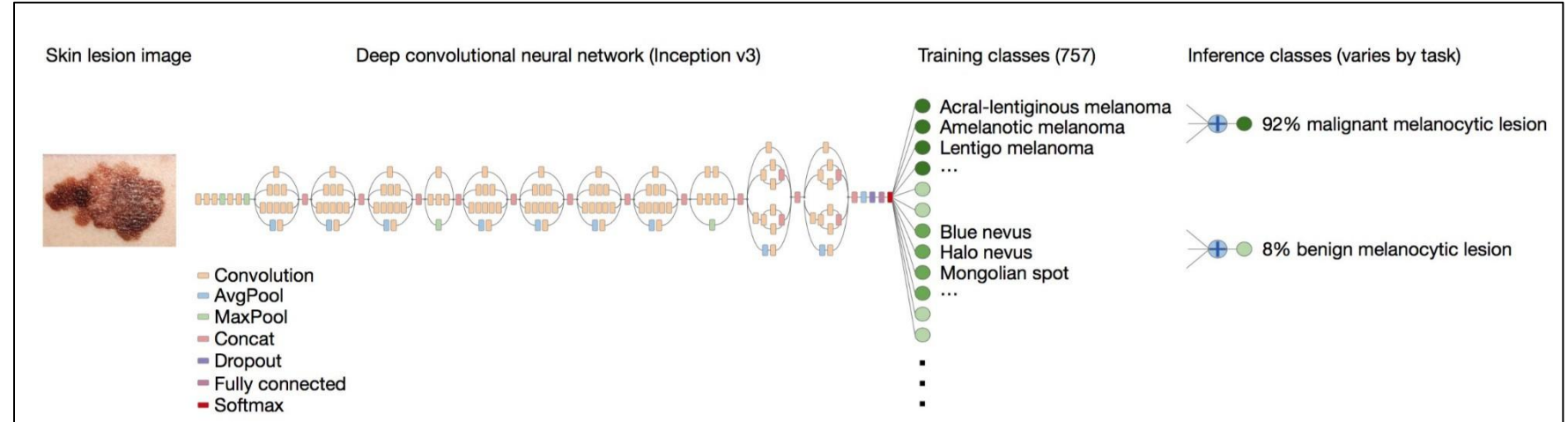
Amazon Scout



# Medical imaging



3D imaging  
(MRI, CT)



Skin cancer classification with deep learning <https://cs.stanford.edu/people/esteva/nature/>





# Current state of the art

- You just saw many examples of current systems.
  - Many of these are less than 5 years old
- Computer vision is an active research area, and rapidly changing
  - Many new apps in the next 5 years
  - Deep learning powering many modern applications
- Many startups across a dizzying array of areas
  - Deep learning, robotics, autonomous vehicles, medical imaging, construction, inspection, VR/AR, ...

# Why is computer vision difficult?



Viewpoint  
variation



Illumination



Credit: Flickr user michaelpaul

Scale

# Why is computer vision difficult?



Intra-class variation



Motion (Source: S. Lazebnik)



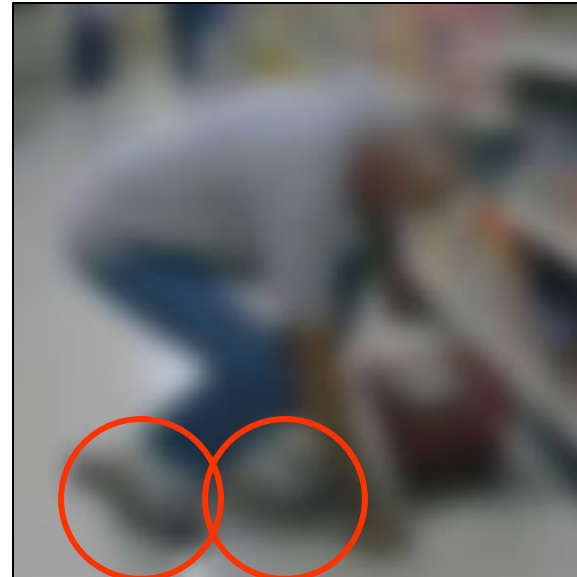
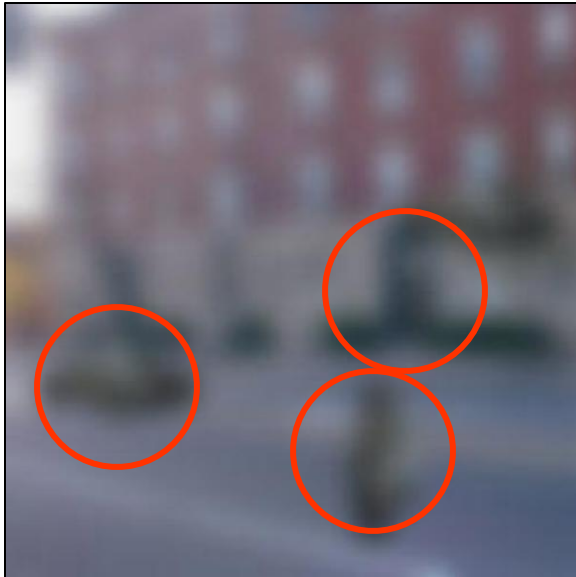
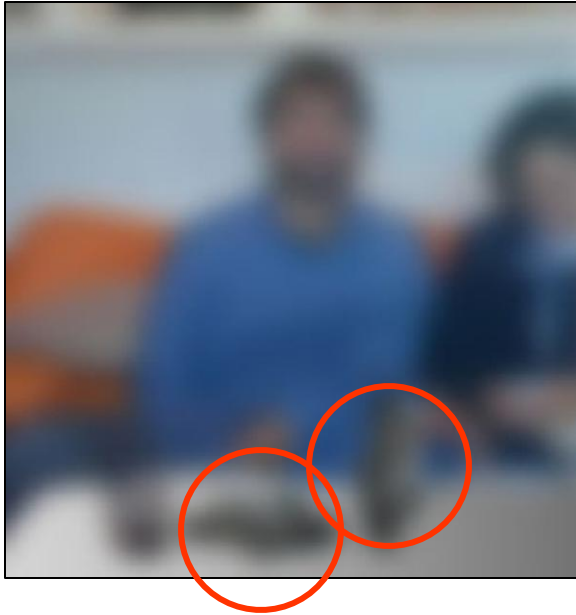
Background clutter



Occlusion



# Challenges: local ambiguity







# But there are lots of visual cues we can use...



Source: S. Lazebnik

# Bottom line

- Perception is an inherently ambiguous problem
  - Many different 3D scenes could have given rise to a given 2D image



Artist Julian Beever with his anamorphic Coke bottle

- We often must use prior knowledge about the world's structure



The picture above is funny.

But for me it is also one of those examples that make me sad about the outlook for AI and for Computer Vision. What would it take for a computer to understand this image as you or I do? I challenge you to think explicitly of all the pieces of knowledge that have to fall in place for it to make sense. Here is my short attempt:

- You recognize it is an image of a bunch of people and you understand they are in a hallway
- You recognize that there are 3 mirrors in the scene so some of those people are "fake" replicas from different viewpoints.
- You recognize Obama from the few pixels that make up his face. It helps that he is in his suit and that he is surrounded by other people with suits.
- You recognize that there's a person standing on a scale, even though the scale occupies only very few white pixels that blend with the background. But, you've used the person's pose and knowledge of how people interact with objects to figure it out.
- You recognize that Obama has his foot positioned just slightly on top of the scale. Notice the language I'm using: It is in terms of the 3D structure of the scene, not the position of the leg in the 2D coordinate system of the image.
- You know how physics works: Obama is leaning in on the scale, which applies a force on it. Scale measures force that is applied on it, that's how it works => it will over-estimate the weight of the person standing on it.
- The person measuring his weight is not aware of Obama doing this. You derive this because you know his pose, you understand that the field of view of a person is finite, and you understand that he is not very likely to sense the slight push of Obama's foot.
- You understand that people are self-conscious about their weight. You also understand that he is reading off the scale measurement, and that shortly the over-estimated weight will confuse him because it will probably be much higher than what he expects. In other words, you reason about implications of the events that are about to unfold seconds after this photo was taken, and especially about the thoughts and how they will develop inside people's heads. You also reason about what pieces of information are available to people.
- There are people in the back who find the person's imminent confusion funny. In other words you are reasoning about state of mind of people, and their view of the state of mind of another person. That's getting frighteningly meta.
- Finally, the fact that the perpetrator here is the president makes it maybe even a little more funnier. You understand what actions are more or less likely to be undertaken by different people based on their status and identity.



# The state of Computer Vision and AI: we are really, really far.

Oct 22, 2012



The picture above is funny.





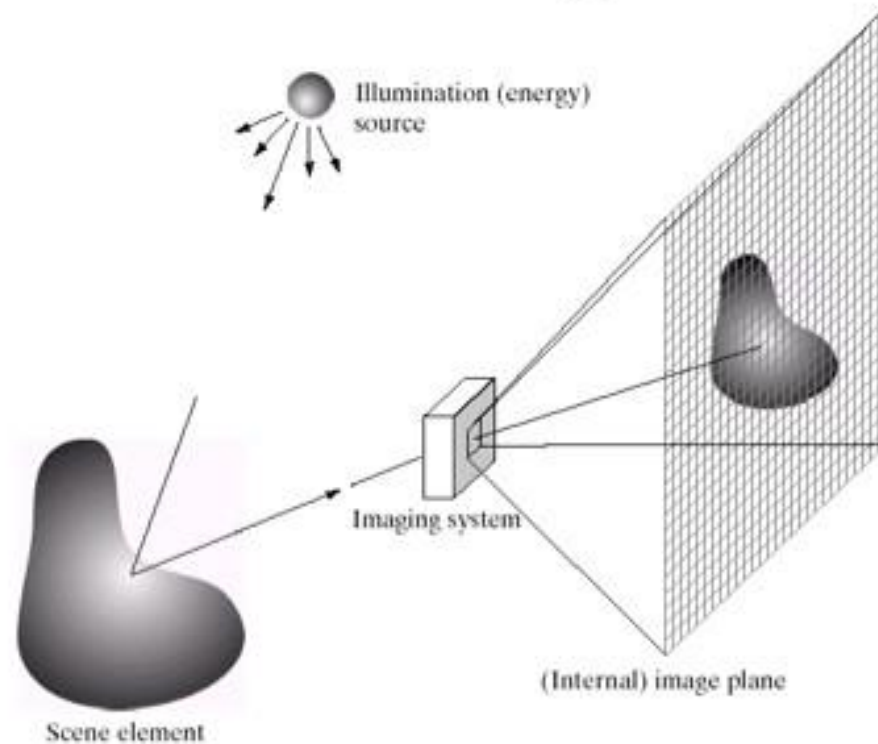
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# What is an image?



# What is an image?



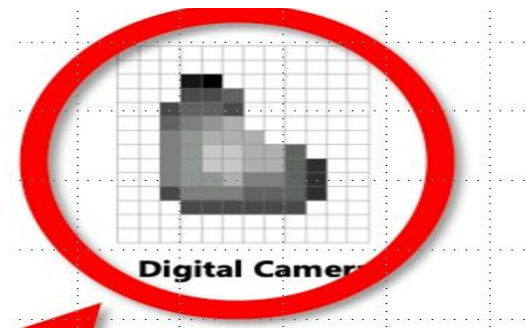
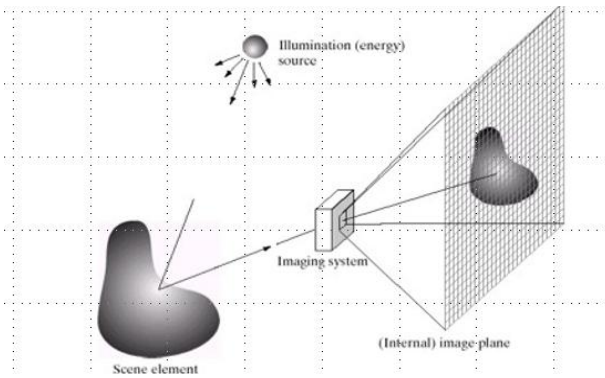
**We'll focus on these in this class**

**(More on this process later)**



Source: A. Efros





- **Illumination (Energy) Source:** This is the light source (like the sun or a light bulb). Without energy hitting an object, there is no reflected light for a camera to "see."
- **Scene Element:** This represents the real-world object. Light from the source hits this object and reflects off its surface toward the camera.
- **Imaging System:** This represents the camera lens and aperture. Its job is to collect the reflected light and project it onto a flat surface.
- The **light passing through the imaging system is projected onto an internal plane**. At this stage, the image is still **continuous**.
- Notice the grid behind the shape; this represents the transition from a smooth, real-world shadow to something that can be measured.
- The lines converging through the "Imaging system" show how a 3D object is mapped onto a 2D surface.

## The Digital Representation (Right Side)

The red circle highlights the final step:

**Digitization.** This is what a digital camera actually produces.

• **Sampling (The Grid):** The image is divided into a grid of discrete squares called **pixels**.

• **Quantization (The Grayscale):** Instead of smooth gradients, each pixel is assigned a specific numerical value representing its brightness.

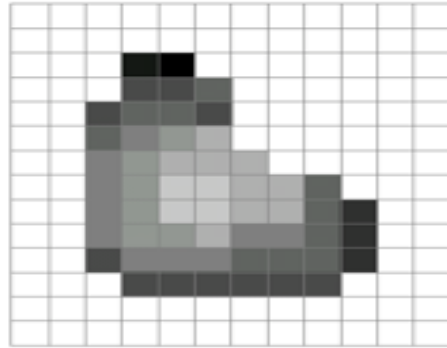
- Darker areas have lower values.
- Lighter areas have higher values.

• **The Result:** What looks like a smooth "blob" in the real world becomes a collection of blocks (pixels) with varying shades of gray when stored in a computer.



# What is an image?

- A grid (matrix) of intensity values



=

|     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 255 | 20  | 0   | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 255 | 75  | 75  | 75  | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 75  | 95  | 95  | 75  | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 96  | 127 | 145 | 175 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 127 | 145 | 175 | 175 | 175 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 127 | 145 | 200 | 200 | 175 | 175 | 95  | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 127 | 145 | 200 | 200 | 175 | 175 | 95  | 47  | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 127 | 145 | 145 | 175 | 127 | 127 | 95  | 47  | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 74  | 127 | 127 | 127 | 95  | 95  | 95  | 47  | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 255 | 74  | 74  | 74  | 74  | 74  | 74  | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |
| 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 | 255 |

(common to use one byte per value: 0 = black, 255 = white)

## Mathematical Perspective

In technical terms, an image is often defined as a two-dimensional function  $f(x, y)$ , where:

- $(x, y)$  are the **spatial coordinates**.
- The value of  $f$  at any point is the **intensity** (brightness) of the image at that point.

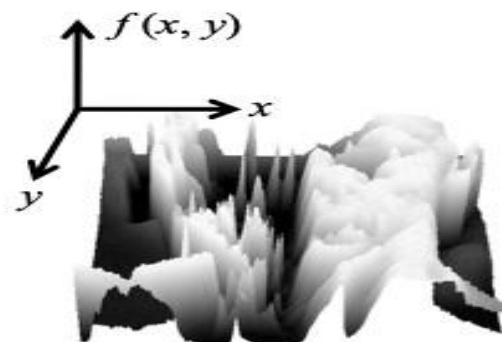
Digital imaging is the process of turning that continuous function into a discrete matrix of numbers that a computer can process.

# What is an image?

- Can think of a (grayscale) image as a **function**  $f$  from  $\mathbb{R}^2$  to  $\mathbb{R}$ :
  - $f(x,y)$  gives the **intensity** at position  $(x,y)$



snoop



3D view

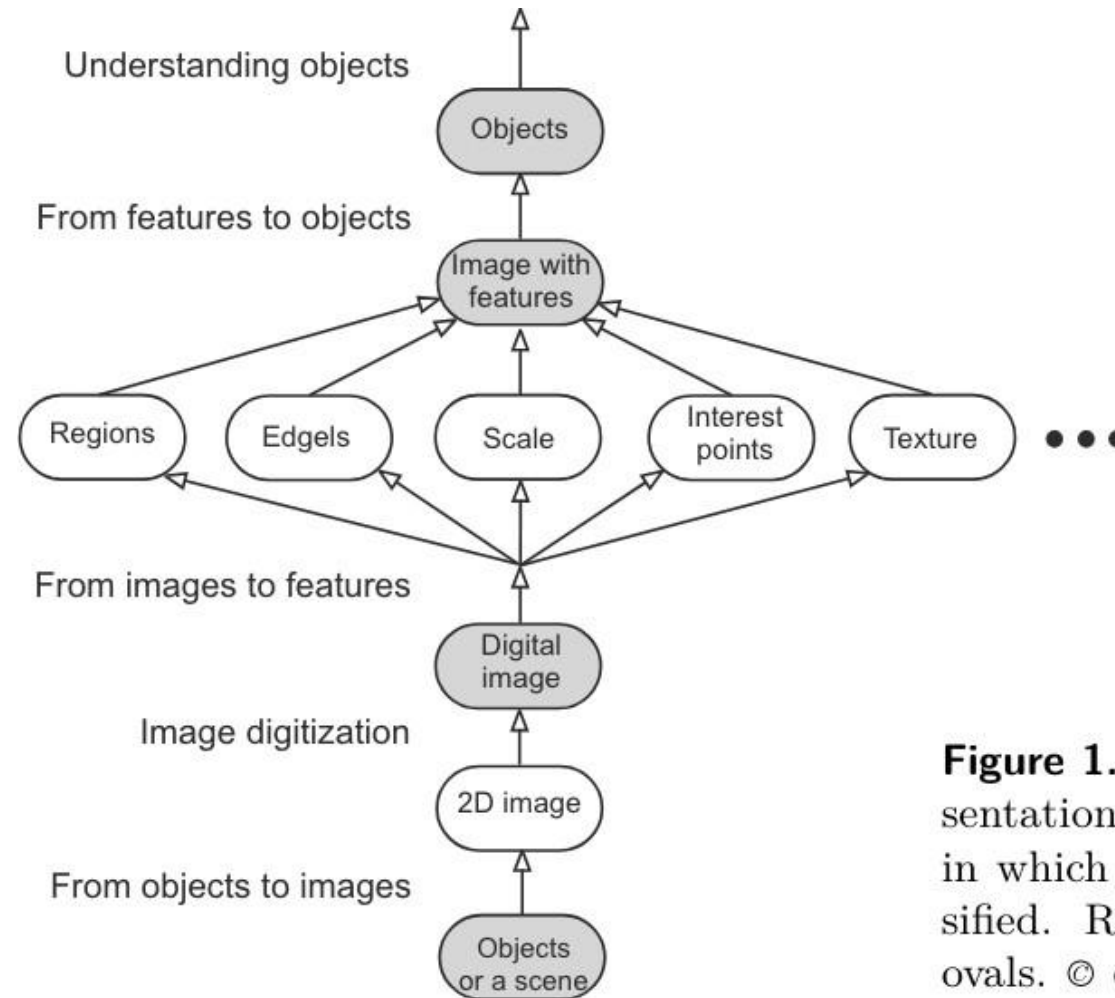
- A **digital** image is a discrete (**sampled, quantized**) version of this function

The slide suggests that we can define a grayscale image as a **mathematical function**.

- **The Function:**  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$
- **The Inputs  $(x, y)$ :** These represent the spatial coordinates (the "address") of a point on the image.
- **The Output  $(f(x, y))$ :** This represents the **intensity** or brightness of the image at that specific point.

| Process             | Description                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Sampling</b>     | Measuring the function at specific, regular intervals (creating <b>pixels</b> ). You go from infinite points to a finite grid. |
| <b>Quantization</b> | Rounding the continuous intensity values to a fixed set of integers (e.g., 0 to 255 for an 8-bit image).                       |

# Image representation for analysis



**Figure 1.7:** Four possible levels of image representation suitable for image analysis problems in which objects have to be detected and classified. Representations are depicted as shaded ovals. © Cengage Learning 2015.

# Explanation

## The Levels of Representation

The process moves from the bottom (physical reality) to the top (semantic understanding):

### Level 1: Objects or a Scene to 2D Image

This is the initial transition from the physical world into a visual format.

### Level 2: Image Digitization

The continuous **2D image** is converted into a **digital image**. This involves the sampling and quantization processes.

### Level 3: From Images to Features

At this stage, the system identifies specific mathematical characteristics within the digital image. Key features identified include:

- **Regions**: Groupings of similar pixels.
- **Edges**: Short for "edge elements," or boundaries.
- **Scale**: The size at which features are viewed.
- **Interest points**: Specific locations with unique characteristics (like corners).
- **Texture**: Repeating patterns or surface properties.

### Level 4: From Features to Objects

The system aggregates these disparate features into a unified **image with features**.

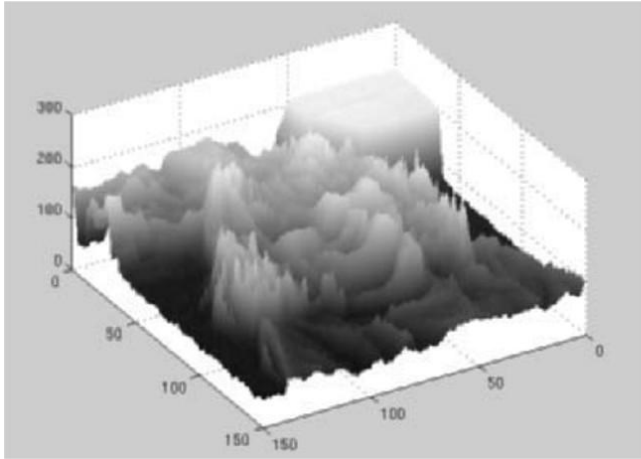
Finally, these features are interpreted to identify and understand distinct **objects** within the scene.

## Core Concept

The main takeaway is that computer vision does not "see" an object immediately. It builds a representation by decomposing an image into basic building blocks (features) and then reconstructing those blocks into a model that matches our human understanding of "objects".



# Representations

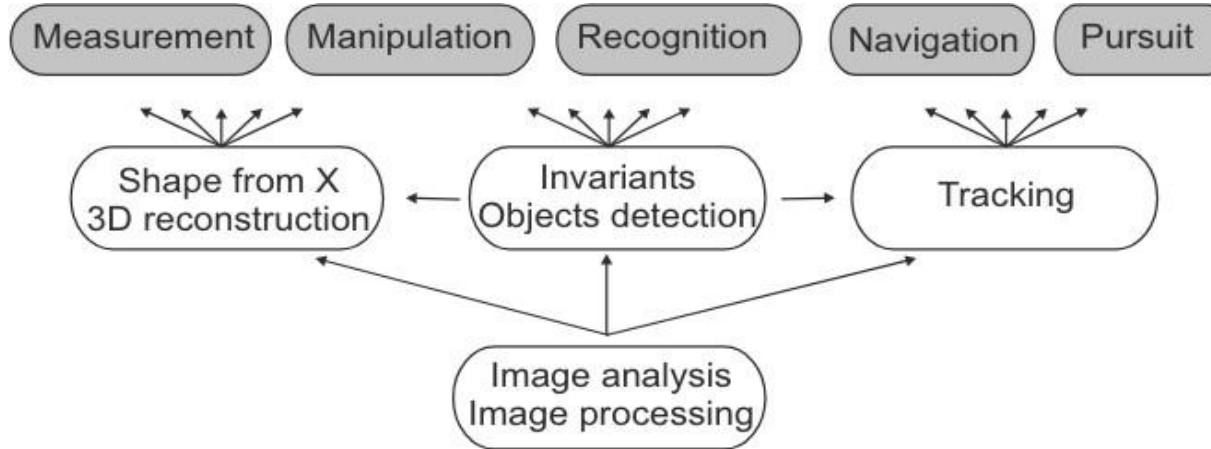


**Figure 1.8:** An unusual image representation. © R.D. Boyle 2015.



**Figure 1.9:** Another representation of Figure 1.8. © R.D. Boyle 2015.

# 3D vision tasks and algorithmic components



## 1. High-Level Vision Tasks (Top Row)

These represent the "end goals" or the practical applications of 3D vision in robotics and AI:

- **Measurement:** Calculating physical dimensions, volumes, or distances in a 3D space.
- **Manipulation:** Enabling a robotic arm or system to interact with and move objects.
- **Recognition:** Identifying specific objects or classes within a 3D environment.
- **Navigation:** Allowing a system to move through space without colliding with obstacles (e.g., self-driving cars).
- **Pursuit:** Following a moving target or maintaining a specific trajectory relative to another object.

## 2. Algorithmic Components (Middle Layer)

- These are the core 3D vision methodologies that bridge the gap between raw pixels and actionable tasks:
- **Shape from X / 3D Reconstruction:** Techniques to recover 3D structure from 2D data (where "X" can be shading, motion, stereo, or texture).
- **Invariants / Object Detection:** Identifying features that remain constant regardless of the camera's viewpoint, which is essential for stable detection.
- **Tracking:** Estimating the state and trajectory of an object over time across a sequence of frames.

## 3. Foundation (Bottom Layer)

**Image Analysis & Image Processing:** This is the bedrock of the entire system. Before 3D can be understood, the system must perform low-level operations like noise reduction, edge detection, filtering, and pixel-level segmentation.

- **Process:** You start with raw images at the bottom.
- **Analyze:** You extract features and detect objects in the middle.
- **Execute:** You use that spatial data to perform the complex tasks at the top.



- Readings:
  1. What is Computer Vision?
  2. Why Computer Vision is hard? (T1 Ch 1.2)
  3. Applications of Computer Vision (R1 Ch 1.1)
  4. Image representation and image analysis tasks (T1 Ch 1.3)
- Topics for Next Class
  5. Image digitization - Sampling and resolution (T1 Ch 2.2)
  6. Digital Images (T1 Ch 2.3)
  7. Digital Image types - Binary, Gray-scale and Color (Class Notes)
  8. Color Images (T1 Ch 2.4)
  9. Color spaces: RGB and HSV (T1 Ch 2.4)

Hands-on session

[https://github.com/mithunkumarsr/LearnComputerVisionWithMithun/blob/main/CV\\_1\\_Understanding\\_Image\\_Representation.ipynb](https://github.com/mithunkumarsr/LearnComputerVisionWithMithun/blob/main/CV_1_Understanding_Image_Representation.ipynb)